

BioCoLoop: PUBLIC-HARNESS COMPARISON

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Paper under double-blind review

Public frameworks use their task-adapted native experimental controllers and the common biological fitting interface. AI-Scientist-v2 and AI-Researcher access laboratory 0, whereas BioCoLoop accesses ten laboratories. Matched Qwen2.5-7B-Instruct and GPT-5.6 Luna blocks use the same twelve-design library, at most six candidate proposals and the same held-out scorer. Scored cells pass independent recomputation from saved predictions; failure tokens record terminal controller outcomes rather than predictive scores.

Model / method	DTI		Proteomics	Cell perturbation		
	BindingDB	PTPC		VCC	Norman	Tahoe
	AUROC $\uparrow$	AP $\uparrow$		Top-1 $\uparrow$	Top-1 $\uparrow$	Top-1 $\uparrow$
Qwen2.5-7B-Instruct						
Fixed model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56		29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00
Direct (1 lab)	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07		21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21
AI-Scientist-v2 (1 lab)	89.68 $\pm$ 0.80	20.18 $\pm$ 0.43		18.61 $\pm$ 2.78	14.12 $\pm$ 0.04	15.42 $\pm$ 6.21
AI-Researcher (1 lab)	86.09 $\pm$ 0.88	20.12 $\pm$ 0.05	19.70 $\pm$ 2.87, F_{tool} (2/3)	21.73 $\pm$ 6.63		17.41 $\pm$ 6.21
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22		31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59
GPT-5.6 Luna (low reasoning)						
Fixed model (1 lab)	82.83 $\pm$ 1.22	25.05 $\pm$ 4.56		29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00
Direct (1 lab)	89.89 $\pm$ 1.02	20.17 $\pm$ 0.42		18.61 $\pm$ 2.78	21.75 $\pm$ 6.59	15.92 $\pm$ 6.03
AI-Scientist-v2 (1 lab)	89.59 $\pm$ 0.95	23.37 $\pm$ 5.89		18.61 $\pm$ 2.78	14.12 $\pm$ 0.04	11.94 $\pm$ 2.11, F_{svc} (2/3)
AI-Researcher (1 lab)	$F_{\text{ctx}}/F_{\text{svc}}$ (0/3)	F_{ctx} (0/3)		F_{ctx} (0/3)	F_{ctx} (0/3)	$F_{\text{ctx}}/F_{\text{svc}}$ (0/3)
BioCoLoop (10 labs)	93.67 $\pm$ 0.98	37.11 $\pm$ 0.01		31.50 $\pm$ 0.00	<u>22.69</u> $\pm$ 0.73	22.39 $\pm$ 2.59

Table 1: Main comparison by research model (mean $\pm$ sample SD, seeds 42–44). Public controllers use laboratory 0 and BioCoLoop uses ten; all rows share the design library, candidate cap, 100-round fits and held-out scorer. Scores are $\times 100$; bold/underline mark the best/second-best complete result per block, and ($n/3$) reports completed seeds without imputing failures.

Outcome. Each model block reports the three-seed mean and sample SD for every complete cell, and preserves the completion count for public-controller failures. Bold and underline identify the best and second-best complete values within each model and endpoint; no incomplete controller cell is ranked.